



Value Proposition: BorrowBox replaces disorganized spreadsheets and informal messaging with a secure, auditable, and automated physical asset sharing workflow for university labs, makerspaces, and shared organizations.

Problem

- **Lost & Unreturned Items:** Informal lending via chat leads to misplaced gear, broken accountability, and lost inventory.
- **Zero Real-Time Visibility:** Members cannot see what items are currently available, who has them, or when they are due.
- **Spreadsheet Bottlenecks:** Manual logs lack access controls, audit trails, and automated overdue tracking.

Solution

- **Live Searchable Catalog:** Real-time status indicators (AVAILABLE, BORROWED, ARCHIVED) with multi-field search.
- **Structured Request Workflow:** Borrowers submit requests; owners review, approve, reject, or confirm loans with due dates.
- **Automated Lifecycle:** Atomic state transitions, overdue calculation, historical logs, and one-click return reconciliation.

Core Features Implemented

MODULE	CAPABILITIES & IMPLEMENTATION DETAILS
Authentication	BCrypt password hashing, stateless HttpOnly JWT cookies (XSS-safe), global exception handling, and React protected route guards.
Inventory Catalog	Full item CRUD, category assignment, dynamic multi-attribute search using Spring Data JPA Specifications, and item archiving.
Borrow Requests	Request submission with custom notes, owner inbox, approve/reject endpoints, and atomic loan confirmation with binding return dates.
Loan Tracking	Active loan monitoring, automated overdue status detection, borrower loan history, and one-click return processing.
User Interface	Clean tabbed dashboard (Explore, My Inventory, Requests, Loans), real-time stats overview, and server-side pagination.

Architecture & Tech Stack

[React 18 SPA (Vite + Nginx)] —(HttpOnly JWT / REST API)—> [Spring Boot 3.5 API (Java 21)] —(JPA / Hibernate)—> [MySQL 8.0]

Backend API

Spring Boot 3.5, Java 21, Spring Security, JPA Specification, Maven

Frontend SPA

React 18, Vite 5, React Router 7, Custom CSS Design System

Database & Auth

MySQL 8.0, HikariCP pool, HttpOnly JWT cookies, BCrypt

DevOps & CI/CD

Docker multi-stage, Nginx, Compose, GitHub Actions CI/CD

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BACKEND TESTS

20

CYPRESS E2E TESTS

100%

DOCKERIZED STACK

CI/CD

AUTOMATED PIPELINES

Future Roadmap (Planned Enhancements)

Multi-Tenant Workspaces & Role Hierarchies

Automated Email & Webhook Overdue Alerts

Mobile QR / Barcode Quick-Checkin

S3 Cloud Item Photo Storage